ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Robotics paper 1 · practice c.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording that shows understanding.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 A ✓ 1.3 C ✓ 1.4 D ✓ 1.5 A ✓ 1.6 B ✓ 1.7 C ✓ 1.8 D ✓ 1.9 B ✓ 1.10 A ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 False ✓ 2.4 True ✓ 2.5 True ✓ (5)

[5]

QUESTION 3

3 3.1 C ✓ 3.2 E ✓ 3.3 G ✓ 3.4 A ✓ 3.5 I ✓ 3.6 J ✓ 3.7 F ✓ 3.8 B ✓ 3.9 H ✓ 3.10 D ✓ (10)

[10]

QUESTION 4

4 4.1 Integrated Development Environment ✓ 4.2 Light-Dependent Resistor ✓ 4.3 Universal Asynchronous Receiver-Transmitter ✓ 4.4 Analogue-to-Digital Converter ✓ 4.5 Lithium Polymer (battery) ✓ (5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 The sensor sends out infrared light and measures how much is reflected ✓; white reflects a lot and black absorbs most of it ✓, so the sensor output changes when it is over the line ✓; the controller uses the readings of the sensors to steer the motors so the line stays under / between the sensors ✓. (4)

5.2 A servo motor turns to a set angle (usually 0–180°) using built-in feedback ✓ – e.g. the joints of a robot arm or a steering mechanism ✓. A stepper motor turns in fixed steps for each pulse, without feedback, and can rotate fully ✓ – e.g. 3D-printer axes, a camera turntable, accurate positioning of a conveyor ✓. (4)

5.3 Motors draw large currents and cause electrical noise and voltage drops ✓ that can reset or damage the Arduino ✓, so they get their own supply. The grounds must be joined so that all signals have the same 0 V reference ✓; otherwise the control signals from the Arduino are not understood by the motor driver ✓. (4)

5.4 Any FOUR in order ✓✓✓✓: connect the Arduino to the computer with a USB cable; open / write the sketch in the Arduino IDE; select the board (Arduino Uno) under Tools; select the correct COM port; click Verify to compile and check for errors; click Upload and wait for "Done uploading". (4)

5.5 Any FOUR ✓✓✓✓: wear safety glasses; work in a well-ventilated area (avoid the fumes); always put the iron back in its stand; never touch the tip or the joint while hot; switch off the iron when not in use; keep the work area dry and tidy; wash hands after handling solder; make sure the circuit is not powered while soldering. (4)

[20]

QUESTION 6

6.1 Torque is the turning force of the motor shaft ✓; speed is how fast it turns, in rpm ✓. Small DC motors spin fast but have little torque ✓, so a gearbox reduces the speed and increases the torque so the robot can move its weight and climb slopes ✓. (4)

6.2 $V_R = 5 - 2 = 3\ V$ ✓; $R = V \div I$ ✓ $= 3 \div 0{,}02$ ✓ $= 150\ \Omega$ ✓ (4)

6.3.1 $I_T = 0{,}6 + 0{,}6 + 0{,}3$ ✓ $= 1{,}5\ A$ ✓ (2)

6.3.2 $P = V \times I = 7{,}4 \times 1{,}5$ ✓ $= 11{,}1\ W$ ✓ (2)

6.3.3 $t = \text{capacity} \div \text{current}$ ✓ $= 2{,}2\ Ah \div 1{,}5\ A$ ✓ $= 1{,}47\ h$ ✓ (about 1 hour 28 minutes) ✓ (4)

[16]

QUESTION 7

7.1 A servo turns to an exact angle that the Arduino sets and holds that position ✓, which is what each joint needs to reach the pick-up and tray positions ✓. (2)

7.2 It measures the colour (red, green and blue light) of the block ✓; the program compares the reading and sends the arm to the red tray or the blue tray ✓. (2)

7.3 It starts the sorting cycle when it is pressed ✓; connected to a digital input pin with a pull-up (INPUT_PULLUP) or pull-down resistor so the input is never floating ✓. (2)

7.4 Any TWO ✓✓: keep hands out of the arm's reach while it moves; test at low speed first; have a way to cut the power quickly (emergency stop / unplug); make sure the arm is clamped down so it cannot tip over; check the wiring before switching on. (2)

7.5 Any ONE sensible improvement with its purpose ✓✓, e.g. an IR / proximity sensor to detect when a block is at the pick-up point so the arm starts by itself; a limit switch to set a home position; a camera to sort more colours or shapes; a force sensor in the gripper so it does not crush the blocks. (2)

[10]

QUESTION 8

8.1 They are PWM pins (marked ~ on the board) ✓✓, so analogWrite() can send a PWM signal to the enable pins ✓ to control the speed of each motor ✓. (4)

8.2 The left motor stops (enA = 0) and the right motor keeps running ✓✓, so the robot turns to the left ✓, back towards the line, until the line is between the sensors again ✓. (4)

8.3 They set the direction pins of the L298N ✓✓ so that both motors turn forward ✓; the speed is then controlled only through the enable pins ✓. (4)

8.4 Both motors stop ✓✓. Useful at a stop line, a junction, the finish line or a charging / loading station ✓✓. (4)

8.5 Line 10 ✓ changes to const int speed = 100; ✓. Duty cycle = 100 ÷ 255 × 100 ✓ = 39,22% ✓ (4)

8.6 digitalRead() returns HIGH (1) or LOW (0) ✓✓. Storing the values once means both decisions in the if statements use the same readings taken at the same moment ✓, and the code is easier to read ✓. (4)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE